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Screen Cover, Foldable Display Module, and Electronic Device

Abstract

A screen cover includes a buffer layer, a first film layer, and a second film layer. The buffer layer absorbs stress of impact through elastic deformation. The first film layer covers a side that is of the buffer layer and that is away from the flexible display, a modulus of the first film layer is greater than a modulus of the buffer layer, and the first film layer is bonded to the buffer layer by using a first adhesive layer. The second film layer covers a side that is of the buffer layer and that is close to the flexible display, a modulus of the second film layer is greater than the modulus of the buffer layer, and the second film layer is bonded to the buffer layer by using a second adhesive layer.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This is a National Stage of International Patent Application No. PCT/CN2022/087379 filed on Apr. 18, 2022, which claims priority to Chinese Patent Application No. 202110437109.1 filed on Apr. 22, 2021 Both of the aforementioned applications are hereby incorporated by reference in their entireties.

TECHNICAL FIELD

[0002] This application relates to the field of electronic device technologies, and in particular, to a screen cover, a foldable display module, and an electronic device.

BACKGROUND

[0003] A flexible foldable display, as a new form of a mobile phone, brings new user experience. However, because a large quantity of flexible polymer materials need to be used in a laminated design (especially a design of a screen cover) of a foldable display module, rigidity and strength of the flexible polymer materials are far lower than those of inorganic materials (for example, cover glass) used in a bar-type phone or a curved-display mobile phone. For example, in Huawei's first-generation outward-folding display screen module, the cover uses a colorless polyimide film with a thickness of 80 μm . Compared with 500 μm cover glass in a bar-type phone, rigidity of the cover is less than 1/2000 of that of the cover glass in the bar-type phone, causing weak resistance to impact and deformation. An accompanying reliability problem of the foldable display module significantly reduces user experience.

SUMMARY

[0004] This application provides a screen cover, a foldable display module, and an electronic device, to improve impact resistance performance of the foldable display module.

[0005] A first aspect of this application provides a screen cover, configured to cover a flexible display. The screen cover includes: a buffer layer, where the buffer layer absorbs stress of impact through elastic deformation; a first film layer, covering a side that is of the buffer layer and that is away from the flexible display, where a modulus of the first film layer is greater than a modulus of the buffer layer, and the first film layer is bonded to the buffer layer by using a first adhesive layer; and a second film layer, covering a side that is of the buffer layer and that is close to the flexible display, where a modulus of the second film layer is greater than the modulus of the buffer layer, and the second film layer is bonded to the buffer layer by using a second adhesive layer.

[0006] The screen cover is configured to cover the flexible display, and can be folded together with the flexible display. The screen cover includes the buffer layer, the first film layer, and the second film layer. The buffer layer absorbs stress of impact through elastic deformation, to reduce tensile strain inside the flexible display. The first film layer covers the side that is of the buffer layer and that is away from the flexible display, that is, the first film layer forms an outer surface of the screen cover. The modulus of the first film layer is greater than the modulus of the buffer layer, to increase strength of the outer surface of the screen cover, and prevent damage such as a scratch on the outer surface of the screen cover. The first film layer is bonded to the buffer layer by using the first adhesive layer. The first adhesive layer is disposed to separate deformation of the first film layer from that of the buffer layer, and the first adhesive layer can generate shear deformation to alleviate conduction of tensile strain on a lower surface of the first film layer to the buffer layer. The second film layer covers the side that is of the buffer layer and that is close to the flexible display, that is, the second film layer is configured to connect to a surface of the flexible display. The modulus of the second film layer is greater than the modulus of the buffer layer, to increase strength of an inner surface of the screen cover, and prevent impact damage on the flexible display

caused by large deformation of the inner surface of the screen cover. The second film layer is bonded to the buffer layer by using the second adhesive layer. The second adhesive layer is disposed to separate deformation of the second film layer from that of the buffer layer, and the second adhesive layer can generate shear deformation to alleviate conduction of tensile strain on a lower surface of the buffer layer to the second film layer.

[0007] Optionally, a thickness of the second film layer is 25 μm to 150 μm , so that the screen cover has good impact resistance performance, and it can be ensured that the screen cover has high folding performance.

[0008] Optionally, a thickness of the second film layer is 25 μm to 75 μm , so that the screen cover has good impact resistance performance, and it can be ensured that the screen cover has excellent folding performance.

[0009] Optionally, the modulus of the second film layer is not less than 1 GPa, to ensure that the second film layer does not generate large deformation after the screen cover is impacted.

[0010] Optionally, a material of the second film layer is colorless polyimide, polyester, epoxy resin, or ultra thin glass.

[0011] Optionally, the second film layer and the first film layer are made of a same material, to simplify a processing technology of the screen cover, and reduce production costs of the screen cover.

[0012] A second aspect of this application provides a foldable display module, including: a flexible display; and a screen cover, covering a surface of the flexible display, where the screen cover is any one of the foregoing screen covers.

[0013] A third aspect of this application provides a foldable display module, including: a flexible display; and a screen cover, covering a surface of the flexible display, where the screen cover includes: a buffer layer, where the buffer layer absorbs stress of impact through elastic deformation; and a first film layer, covering a side that is of the buffer layer and that is away from the flexible display, where a modulus of the first film layer is greater than a modulus of the buffer layer, and the first film layer is bonded to the buffer layer by using a first adhesive layer.

[0014] The buffer layer is bonded to the surface of the flexible display by using a third adhesive layer, and a thickness of the third adhesive layer is not less than half of a thickness of the buffer layer.

[0015] The foldable display module includes the flexible display and the screen cover. The screen cover covers the surface of the flexible display, and the screen cover includes the buffer layer and the first film layer. The buffer layer absorbs stress of impact through elastic deformation, to reduce tensile strain inside the flexible display. The first film layer covers the side that is of the buffer layer and that is away from the flexible display, that is, the first film layer forms an outer surface of the screen cover. The modulus of the first film layer is greater than the modulus of the buffer layer, to increase strength of the outer surface of the screen cover, and prevent damage such as a scratch on the outer surface of the screen cover. The first film layer is bonded to the buffer layer by using the first adhesive layer. The first adhesive layer is disposed to separate deformation of the first film layer from that of the buffer layer, and prevent strain on the buffer layer from conducting to the first film layer. The buffer layer is bonded to the surface of the flexible display by using the third adhesive layer, and the thickness of the third adhesive layer is not less than half of the thickness of the buffer layer. The third adhesive layer with a large thickness is used to isolate the buffer layer from the flexible display, thereby alleviating conduction of strain on the buffer layer to the flexible display.

[0016] Optionally, the thickness of the buffer layer is not greater than 50 μm , and the thickness of the third adhesive layer is 15 μm to 75 μm .

[0017] Optionally, the thickness of the buffer layer is 50 μm to 100 μm , and the thickness of the third adhesive layer is 25 μm to 100 μm .

[0018] Optionally, the thickness of the buffer layer is 100 μm to 150 μm , and the thickness of the

third adhesive layer is 50 μm to 125 μm .

[0019] Optionally, the thickness of the buffer layer is 150 μm to 200 μm , and the thickness of the third adhesive layer is 50 μm to 150 μm .

[0020] Optionally, the thickness of the buffer layer is not less than 200 μm , and the thickness of the third adhesive layer is not less than 75 μm .

[0021] A fourth aspect of this application provides an electronic device, including any one of the foregoing foldable display modules.

[0022] It should be understood that the foregoing general description and the following detailed description are merely examples, and cannot limit this application.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0023] FIG. 1 is a schematic diagram of a structure of a foldable display module in the conventional technology;

[0024] FIG. 2 is a schematic diagram of a structure of another foldable display module in the conventional technology;

[0025] FIG. 3 is a schematic diagram of an impact test with a falling ball;

[0026] FIG. 4 is a schematic diagram of a structure of a first foldable display module according to an embodiment of this application;

[0027] FIG. 5 is a status diagram of deformation of a foldable display module in an impact scenario;

[0028] FIG. 6 shows force analysis of transverse deformation occurring in FIG. 5;

[0029] FIG. 7 is a schematic diagram of bending deformation occurring in FIG. 5;

[0030] FIG. 8 is a schematic diagram of a structure of a second foldable display module according to an embodiment of this application; and

[0031] FIG. 9 is a schematic diagram of a structure of a third foldable display module according to an embodiment of this application.

REFERENCE NUMERALS

[0032] **01**: plane area; [0033] **02**: bending area; [0034] **03**: flexible filler; [0035] **1**: screen cover; [0036] **11**: buffer layer; [0037] **12**: first film layer; [0038] **13**: first adhesive layer; [0039] **14**: second film layer; [0040] **15**: second adhesive layer; [0041] **16**: third adhesive layer; [0042] **2**: flexible display; [0043] **21**: polarizer; [0044] **22**: panel; and [0045] **3**: support assembly.

[0046] The accompanying drawings herein are incorporated into the specification and constitute a part of the specification, show embodiments conforming to this application, and are used together with the specification to explain a principle of this application.

DESCRIPTION OF EMBODIMENTS

[0047] For ease of understanding technical solutions of this application, the following describes embodiments of this application in detail with reference to accompanying drawings.

[0048] Terms used in embodiments of this application are merely for the purpose of describing specific embodiments, but are not intended to limit this application. The terms “a”, “the” and “this” of singular forms used in embodiments and the appended claims of this application are also intended to include plural forms, unless otherwise specified in the context clearly.

[0049] It should be understood that the term “and/or” in this specification describes only an association relationship of associated objects and represents that three relationships may exist. For example, A and/or B may represent the following three cases: Only A exists, both A and B exist, and only B exists. In addition, the character “/” in this specification generally indicates an “or” relationship between the associated objects.

[0050] It should be noted that orientation words such as “above”, “below”, “left”, and “right”

described in embodiments of this application are described from perspectives shown in the accompanying drawings, and shall not be understood as a limitation on embodiments of this application. Moreover, in the context, it also should be understood that, when it is mentioned that one element is connected “above” or “below” another element, the element can be directly connected “above” or “below” the another element, or may be indirectly connected “above” or “below” the another element through an intermediate element.

[0051] Embodiments of this application provide an electronic device, including a foldable display module. The electronic device may be a common mobile terminal like a mobile phone or a tablet computer, and the electronic device may be in a folded state or an unfolded state. In the folded state, the foldable display module is bent together with the electronic device. In the unfolded state, the foldable display module is unfolded together with the electronic device into a plane.

[0052] The foldable display module provided in embodiments of this application includes a screen cover **1**, a flexible display **2**, and a support assembly **3**. The flexible display **2** may be an integrated flexible display screen, or may be a display screen formed by splicing two rigid displays and one flexible display located between the two rigid displays. The support assembly **3** supports the flexible display **2**, and the screen cover **1** covers a side that is of the flexible display **2** and that is away from the support assembly **3**. In other words, the screen cover **1** is attached to an outer surface of the flexible display **2**, and the screen cover **1** protects the flexible display **2** and improves surface flatness of the flexible display **2**. The screen cover **1** has light transmission, and an image displayed on the flexible display **2** may be seen through the screen cover **1**. In addition, the screen cover **1** can be bent or flattened synchronously with the flexible display **2**, to adapt to switching between an unfolded state and a folded state of the foldable display module.

[0053] As a protective layer of a screen, a screen cover plays a most important role in an impact resistance capability of a module. Therefore, structural optimization of the screen cover is naturally one of important improvement directions for improving impact resistance performance of a foldable display module. Refer to FIG. 1 and FIG. 2. A screen cover mainly includes a plane area **01** and a bending area **02**. The bending area **02** is located in the middle, and the plane area **01** is distributed on two sides of the bending area **02**. The plane area **01** does not need to be bent or deformed, that is, the plane area **01** always maintains a plane state. The bending area **02** may be bent and flattened, so that the screen cover is folded and unfolded together with a flexible display.

[0054] As shown in FIG. 1, thick cover glass (100 μm to 200 μm) is used in the plane area **01**, and a thickness of the bending area **02** is less than that of the plane area **01**. That is, partial thinning is performed on the bending area **02**, and the partially thinned part is filled by a flexible filler **03**. The flexible filler **03** is a low modulus polymer material. To implement a bending characteristic of glass, a thickness of the glass generally needs to be controlled within 70 μm , that is, a thickness of the bending area **02** does not exceed 70 μm . According to a current partial thinning technology, the thick glass used is generally 100 μm to 200 μm , that is, a thickness of the plane area **01** is 100 μm to 200 μm . Compared with the 500 μm cover glass in a bar-type phone, the plane area **01** differs greatly in rigidity. Therefore, it cannot be ensured that impact resistance performance of the plane area **01** is consistent with that of the bar-type phone, and the bending area **02** has a weaker impact resistance capability due to the thinner glass. It is difficult for the screen cover with such a structure to meet a requirement of impact resistance performance.

[0055] As shown in FIG. 2, thick cover glass (100 μm to 500 μm) is used in the plane area **01**, and a hole area is disposed on the bending area **02**. That is, partial etching (a through-hole may be etched or a blind hole may be etched) is performed on the bending area **02**, and the etched hole area is filled by a flexible filler **03**. The flexible filler **03** is a low modulus polymer material. In this solution, because partial etching is performed on the bending area **02**, the bending area **02** has a regional section difference of a hole area location, and a stress concentration point is formed on the screen when impact occurs, causing poor impact resistance reliability of the bending area.

[0056] Due to insufficient impact resistance reliability of the screen covers of the foregoing two

structure types, the foldable display module is prone to a problem of broken bright spots. According to the second failure criterion for brittle materials, a main cause of broken bright spots on a screen is that when a foldable display module is affected by external impact, a thin film transistor (Thin film transistor, TFT) in a panel structure is subject to impact compressive stress, and a part of the thin film transistor is bent and deformed. As a result, tensile strain on the TFT exceeds limit strain, and a short circuit or an open circuit occurs in a related area, causing display exceptions such as broken bright spots. Refer to FIG. 3. Impact resistance performance of a foldable display module may be evaluated through an impact test with a falling ball, that is, a steel ball of a specified weight (for example, 32 g) freely falls onto a product from a specified falling height, to impact a product, and then appearance and various aspects of performance of the product are checked. In a current laminated structure solution, foldable display modules perform extremely poorly in tests of falling-ball impact resistance. For example, the falling-ball impact resistance capability of a module of the Mate Xs is only 4.5 cm for the plane area and 2.8 cm for the bending area, which cannot meet basic requirements of users for impact resistance performance in daily use. [0057] In addition, the cover glass in both the foregoing two structures needs to be filled by the flexible filler **03**. If no trace is left in optical terms, it needs to be ensured that refractive indexes of the glass and the polymer material are highly matched (generally, six decimal places of the refractive indexes of the two materials need to be consistent). However, because a molecular weight of the polymer material is generally distributed in a form of normal distribution, precise control of the refractive index cannot be implemented. Therefore, a requirement of no light shadow cannot be met in optical terms, affecting appearance of a foldable display module.

[0058] For a foldable display module provided in Embodiment **1** of this application, refer to FIG. 4. The foldable display module includes a screen cover **1**, a flexible display **2**, and a support assembly **3**. The flexible display **2** includes a polarizer (Polarizer, POL) **21** and a panel **22**. The polarizer **21** is disposed on a side that is of the panel **22** and that faces the screen cover **1**. A TFT is disposed in the panel **22**. The screen cover **1** is configured to cover the flexible display **2**, and can be folded together with the flexible display **2**. The screen cover **1** is bonded to a surface of the flexible display **2** by using a third adhesive layer **16** (for example, OCA). The screen cover **1** includes at least two layers of structures, and the layers are sequentially stacked to form a laminated structure design, so that the screen cover **1** forms a continuous surface structure, to improve appearance of the foldable display module. The screen cover **1** includes at least one buffer layer **11**. For example, the screen cover **1** may include a buffer layer **11** and a first film layer **12**. The first film layer **12** covers a side that is of the buffer layer **11** and that is away from the flexible display **2**, that is, the first film layer **12** forms an outer surface of the screen cover **1**. A transparent buffer layer **11** with an appropriate modulus is selected as the buffer layer **11**. That is, the buffer layer **11** absorbs stress of impact through elastic deformation, to protect a screen and greatly improve impact resistance performance of the module laminate. A modulus of the first film layer **12** is greater than a modulus of the buffer layer **11**, to increase strength of the outer surface of the screen cover **1**, and prevent damage such as a scratch on the outer surface of the screen cover **1**. The first film layer **12** is bonded to the buffer layer **11** by using a first adhesive layer **13**. The first adhesive layer **13** is disposed to separate deformation of the first film layer **12** from that of the buffer layer **11**, and the first adhesive layer **13** can generate shear deformation to alleviate conduction of tensile strain on a lower surface of the first film layer **12** to the buffer layer **11**.

[0059] The buffer layer **11** is made of a transparent material whose modulus is 10 Mpa to 200 Mpa. For example, the modulus of the buffer layer **11** may be 10 Mpa, 30 Mpa, 50 Mpa, 60 Mpa, 80 Mpa, 100 Mpa, 110 Mpa, 130 Mpa, 150 Mpa, 160 Mpa, 180 Mpa, 200 Mpa, or the like. In an optional implementation, the buffer layer **11** is made of a thermoplastic PU (Thermoplastic PU, TPU) material.

[0060] The first film layer **12** is made of a transparent material with a high modulus, and the modulus of the first film layer **12** is far greater than the modulus of the buffer layer **11**. The first

film layer **12** needs to meet characteristics of being bendable and having high surface strength. For example, the first film layer **12** may be made of a polyester (Polyethylene terephthalate, PET) material. The first adhesive layer **13** may be an optical clear adhesive (Optical clear adhesive, OCA). A modulus of the first adhesive layer **13** is far less than the modulus of the buffer layer **11**. When the buffer layer **11** is deformed due to impact, the first adhesive layer **13** can generate shear deformation to store energy, so that the buffer layer **11** and the first film layer **12** can form respective neutral layers, and the deformation of the buffer layer **11** does not affect the first film layer **12**.

[0061] To describe impact resistance performance of the screen cover **1** provided in this embodiment, a comparison experiment is performed on impact resistance performance of common foldable display modules and impact resistance performance of foldable display modules that use the screen cover provided in this embodiment. For results of the comparison experiment, refer to Table **1**. The specific experimental manner is an impact test with a falling ball (refer to FIG. **3**). Except for a structure of the screen cover **1**, other parameters of the foldable display modules according to this embodiment in each group are the same, and other experimental conditions are also the same. Each experimental group represents a group of sample foldable display modules, and an experimental result of the group is an average value of experimental results of the group of sample foldable display modules.

TABLE-US-00001 TABLE 1 Group Experimental Experimental Experimental Control group 1															
group 1	group 2	group 3	Structure	Material	Thickness	Material	Thickness	Material	Thickness	Material	Thickness	Material	Thickness	Material	Thickness
Cover	PET	50	PET	50	PET	50	PET	50	OCA	50	OCA	50	OCA	50	PET
100	TPU	100	TPU	150	TPU	200	OCA	50	OCA	50	OCA	50	OCA	50	Impact
2 cm ok	18 cm ok	16 cm ok	with a 32-3 cm broken	16 cm broken	18 cm broken	20 cm broken	g	falling	bright	spots	bright	spots	bright	spots	ball

[0062] It may be learned from the experimental data in Table **1** that, Control group **1** uses a PET layer without buffer performance, and a height of falling-ball impact resistance of the module is 2 cm. Experimental group **1** to Experimental group **3** use TPU layers with buffer effect, and heights of falling-ball impact resistance of the modules are 14 cm, 16 cm, and 18 cm respectively. Therefore, in the screen cover **1** provided in this embodiment of this application, the buffer layer **11** is disposed to effectively reduce stress of impact, and reduce tensile strain on the TFT layer caused by the stress of impact, thereby improving impact resistance performance of the foldable display module. In addition, as a thickness of the buffer layer **11** increases, impact resistance performance of the screen cover **1** gradually improves. However, it may be learned from comparison between Experimental group **1** to Experimental group **3** that, when a thickness of a buffer layer (TPU) is increased from 100 μm to 200 μm , the thickness of the buffer layer is doubled, but impact resistance performance of the foldable display module is not improved accordingly. For example, when a 100 μm buffer layer is used, a height of falling-ball impact resistance of the module is 14 cm, but when the thickness of the buffer layer is increased from 100 μm to 200 μm , a height of falling-ball impact resistance of the module is 18 cm. The height of falling-ball impact resistance of the module is increased by only 4 cm.

[0063] Refer to FIG. **5** to FIG. **7**. It may be further learned through analysis that, in an impact scenario, there are two sources for the TFT layer to generate tensile strain. One is shown in FIG. **6**. That is, the TFT layer generates transverse deformation under longitudinal impact compressive stress. The other is shown in FIG. **7**. Tensile strain is generated due to partial bending of the laminated structure of the foldable display module in the impact process, and dashed lines represent the neutral layer of the buffer layer **11**.

[0064] It may be learned from FIG. **6** that, the buffer layer **11** can reduce longitudinal stress of impact, and reduce transverse deformation of the TFT layer under the stress of impact. Therefore, impact resistance performance of the screen cover **1** can be improved by disposing the buffer layer **11**. It may be learned from FIG. **7** that, along a direction indicated by an arrow, a thickness of the

buffer layer **11** gradually increases, and a distance between a lower surface of the buffer layer **11** and the neutral layer of the buffer layer **11** becomes longer, and tensile strain on the lower surface of the buffer layer **11** becomes larger. Because the lower surface of the buffer layer **11** is close to the TFT layer in the panel, the tensile strain on the lower surface of the buffer layer **11** may conduct to the TFT layer in the panel, causing the tensile strain on the TFT layer in the panel to increase. Therefore, when the buffer layer **11** is thickened, tensile stress on the lower surface of the buffer layer **11** increases, and when the buffer layer **11** is close to the panel, affected by the tensile strain on the lower surface of the buffer layer **11**, the tensile strain on the TFT layer increases. Consequently, an increase in the thickness of the buffer layer **11** cannot significantly improve impact resistance performance of the screen cover **1**.

[0065] For a foldable display module provided in Embodiment **2** of this application, refer to FIG. **8**. The foldable display module includes a screen cover **1**, a flexible display **2**, and a support assembly **3**. The screen cover **1** covers a surface of the flexible display **2**, and the screen cover **1** includes a buffer layer **11** and a first film layer **12**. The buffer layer **11** absorbs stress of impact through elastic deformation, to reduce tensile strain inside the flexible display **2**. The first film layer **12** covers a side that is of the buffer layer **11** and that is away from the flexible display **2**, that is, the first film layer **12** forms an outer surface of the screen cover **1**. A modulus of the first film layer **12** is greater than a modulus of the buffer layer **11**, to increase strength of the outer surface of the screen cover **1**, and prevent damage such as a scratch on the outer surface of the screen cover **1**. The first film layer **12** is bonded to the buffer layer **11** by using a first adhesive layer **13**. The first adhesive layer **13** is disposed to separate deformation of the first film layer **12** from that of the buffer layer **11**, and the first adhesive layer **13** can generate shear deformation to alleviate conduction of tensile strain on a lower surface of the first film layer **12** to the buffer layer **11**. The buffer layer **11** is bonded to the surface of the flexible display **2** by using a third adhesive layer **16**, and a thickness of the third adhesive layer **16** is not less than half of a thickness of the buffer layer **11**. The third adhesive layer **16** with a large thickness is used to isolate the buffer layer **11** from the flexible display **2**, and the third adhesive layer **16** can generate shear deformation to effectively alleviate conduction of strain on the buffer layer **11** to the flexible display **2**.

[0066] Specifically, a modulus of the third adhesive layer **16** is far less than the modulus of the buffer layer **11**. When the thickness of the buffer layer **11** is increased, the thickness of the third adhesive layer **16** (for example, an OCA) on a lower surface of the buffer layer **11** is also increased. That is, the thickness of the OCA on the lower surface of the buffer layer **11** needs to match the thickness of the buffer layer **11**. A distance between the lower surface of the buffer layer **11** and the flexible display **2** may be increased by increasing the thickness of the third adhesive layer **16**, so that it is difficult for the strain on the buffer layer **11** to conduct to a TFT layer. In addition, because a modulus of the OCA is generally 10 Kpa to 100 Kpa, which is far less than the modulus (10 Mpa to 200 Mpa) of the buffer layer **11**, when the module is partially bent in an impact process, the OCA may generate shear deformation, to store strain on the lower surface of the buffer layer **11** in a form of shear strain energy, thereby alleviating downward conduction of the strain. Specific solutions are as follows:

[0067] (1) The thickness of the buffer layer **11** is not greater than 50 μm , and the thickness of the third adhesive layer **16** is 15 μm to 75 μm . That is, when the thickness of the buffer layer **11** is less than or equal to 50 μm , the thickness of the OCA is preferably 15 μm to 75 μm . Typically but not as a limitation, when the thickness of the buffer layer **11** is 25 μm , the thickness of the OCA is preferably 15 μm to 25 μm , for example, the thickness of the OCA is 15 μm or 25 μm . When the thickness of the buffer layer **11** is 50 μm , the thickness of the OCA is preferably 25 μm to 50 μm , for example, the thickness of the OCA is 25 μm or 50 μm .

[0068] (2) The thickness of the buffer layer **11** is 50 μm to 100 μm , and the thickness of the third adhesive layer **16** is 25 μm to 100 μm . That is, when the thickness of the buffer layer **11** is between 50 μm and 100 μm , the thickness of the OCA is preferably 25 μm to 100 μm . Typically but not as a

limitation, when the thickness of the buffer layer **11** is 100 μm , the thickness of the OCA is preferably 50 μm to 75 μm , for example, the thickness of the OCA is 50 μm or 75 μm .

[0069] (3) The thickness of the buffer layer **11** is 100 μm to 150 μm , and the thickness of the third adhesive layer **16** is 50 μm to 125 μm . That is, when the thickness of the buffer layer **11** is between 100 μm and 150 μm , the thickness of the OCA is preferably 50 μm to 125 μm . Typically but not as a limitation, when the thickness of the buffer layer **11** is 150 μm , the thickness of the OCA is preferably 75 μm to 100 μm , for example, the thickness of the OCA is 75 μm or 100 μm .

[0070] (4) The thickness of the buffer layer **11** is 150 μm to 200 μm , and the thickness of the third adhesive layer **16** is 50 μm to 150 μm . That is, when the thickness of the buffer layer **11** is between 150 μm and 200 μm , the thickness of the OCA is preferably 50 μm to 150 μm . Typically but not as a limitation, when the thickness of the buffer layer **11** is 175 μm , the thickness of the OCA is preferably 75 μm to 125 μm , for example, the thickness of the OCA is 75 μm , 100 μm , or 125 μm .

[0071] (5) The thickness of the buffer layer **11** is not less than 200 μm , and the thickness of the third adhesive layer **16** is not less than 75 μm . That is, when the thickness of the buffer layer **11** is greater than or equal to 200 μm , the thickness of the OCA is preferably greater than or equal to 75 μm .

[0072] To describe impact resistance performance of the screen cover **1** provided in this embodiment, a comparison experiment is performed on impact resistance performance of foldable display modules with different thicknesses of the third adhesive layer **16**. For results of the comparison experiment, refer to Table 2. The specific experimental manner is an impact test with a falling ball (refer to FIG. 3). Except for a structure of the screen cover **1**, other parameters of the foldable display modules according to this embodiment in each group are the same, and other experimental conditions are also the same. Each experimental group represents a group of sample foldable display modules, and an experimental result of the group is an average value of experimental results of the group of sample foldable display modules.

TABLE-US-00002

TABLE 2	Group	Experimental group 11	Experimental group 12	Experimental group 13	Structure	Material	Thickness	Material	Thickness	Material	Thickness	Cover	PET	50	PET	50	PET	50	OCA	50	OCA	50	OCA	50	TPU	150	TPU	150	TPU	150	OCA	50	OCA	100	OCA	150
Impact	16	cm	ok	20	cm	ok	24	cm	ok	with a	32-	18	cm	broken	22	cm	broken	26	cm	broken	g	falling	bright	spots	bright	spots	bright	spots	ball							

[0073] It may be learned from the experimental data in Table 2 that, impact resistance performance of the foldable display module can be effectively improved by increasing the thickness of the third adhesive layer **16**.

[0074] For a foldable display module provided in Embodiment 3 of this application, refer to FIG. 9. The foldable display module includes a screen cover **1**, a flexible display **2**, and a support assembly **3**. The screen cover **1** is configured to cover the flexible display **2**, and can be folded together with the flexible display **2**. The screen cover **1** includes a buffer layer **11**, a first film layer **12**, and a second film layer **14**. The buffer layer **11** absorbs stress of impact through elastic deformation, to reduce tensile strain inside the flexible display **2**. The first film layer **12** covers a side that is of the buffer layer **11** and that is away from the flexible display **2**, that is, the first film layer **12** forms an outer surface of the screen cover **1**. A modulus of the first film layer **12** is greater than a modulus of the buffer layer **11**, to increase strength of the outer surface of the screen cover **1**, and prevent damage such as a scratch on the outer surface of the screen cover **1**. The first film layer **12** is bonded to the buffer layer **11** by using a first adhesive layer **13**. The first adhesive layer **13** is disposed to separate deformation of the first film layer **12** from that of the buffer layer **11**, and the first adhesive layer **13** can generate shear deformation to alleviate conduction of tensile strain on a lower surface of the first film layer **12** to the buffer layer **11**. The second film layer **14** covers a side that is of the buffer layer **11** and that is close to the flexible display **2**, that is, the second film layer **14** is configured to connect to a surface of the flexible display **2**. A modulus of the second film layer **14** is greater than the modulus of the buffer layer **11**, to increase strength of an inner surface

of the screen cover **1**, and prevent impact damage on the flexible display **2** caused by large deformation of the inner surface of the screen cover **1**. The second film layer **14** is bonded to the buffer layer **11** by using a second adhesive layer **15**. The second adhesive layer **15** is disposed to separate deformation of the second film layer **14** from that of the buffer layer **11**, and the second adhesive layer **15** can generate shear deformation to alleviate conduction of tensile strain on a lower surface of the buffer layer **11** to the second film layer **14**.

[0075] The second film layer **14** is made of a transparent material with a high modulus, and the modulus of the second film layer **14** is far greater than the modulus of the buffer layer **11**. The second film layer **14** needs to meet characteristics of being bendable and having high surface strength. The modulus and a thickness of the second film layer **14** are controlled to prevent downward conduction of strain. The second adhesive layer **15** may be an optical clear adhesive (Optical clear adhesive, OCA). A modulus of the first adhesive layer **13** is far less than the modulus of the buffer layer **11**. When the buffer layer **11** is deformed due to impact, the first adhesive layer **13** can generate shear deformation to store energy, so that the buffer layer **11** and the second film layer **14** can form respective neutral layers, and the deformation of the buffer layer **11** does not affect the second film layer **14**.

[0076] Further, the thickness of the second film layer **14** is 25 μm to 150 μm . For example, the thickness of the second film layer **14** may be 25 μm , 35 μm , 45 μm , 55 μm , 60 μm , 65 μm , 75 μm , 85 μm , 95 μm , 100 μm , 110 μm , 120 μm , 130 μm , 140 μm , 150 μm , or the like, so that the screen cover **1** has good impact resistance performance, and it can be ensured that the screen cover **1** has high folding performance. When the thickness of the second film layer **14** is less than 25 μm , the second film layer **14** is excessively thin. As a result, the deformation of the buffer layer **11** may conduct to the flexible display **2**, and it is difficult for the impact resistance performance of the foldable display module to meet a use requirement. When the thickness of the second film layer **14** is greater than 150 μm , the second film layer **14** is excessively thick. Because the second film layer **14** is a high modulus material, it is inconvenient to fold the second film layer **14**. In addition, the excessively thick second film layer **14** causes an increase in an overall thickness of the foldable display module, thereby affecting folding performance of the foldable display module.

[0077] Preferably, the thickness of the second film layer **14** is 25 μm to 75 μm . For example, the thickness of the second film layer **14** may be 25 μm , 30 μm , 35 μm , 40 μm , 45 μm , 50 μm , 55 μm , 60 μm , 65 μm , 70 μm , 75 μm , or the like, so that the screen cover **1** has good impact resistance performance, and it can be ensured that the screen cover **1** has excellent folding performance.

[0078] Further, the modulus of the second film layer **14** shall be not less than 1 Gpa, to ensure that the second film layer **14** does not generate large deformation after the screen cover **1** is impacted, thereby preventing the deformation of the buffer layer **11** from conducting to the flexible display **2**.

[0079] Further, a material of the second film layer **14** may be colorless polyimide (Colorless polyimide, CPI), polyester (Polyethylene terephthalate, PET), epoxy resin (Epoxy resin, EP), ultra thin glass (Ultra thin glass, UTG), or the like.

[0080] Further, the second film layer **14** and the first film layer **12** are made of a same material, to simplify a processing technology of the screen cover **1**, and reduce production costs of the screen cover **1**.

[0081] To describe impact resistance performance of the screen cover **1** provided in this embodiment, a comparison experiment is performed on impact resistance performance of foldable display modules having screen covers with different structures. For results of the comparison experiment, refer to Table 3. The specific experimental manner is an impact test with a falling ball (refer to FIG. 3). Except for a structure of the screen cover **1**, other parameters of the foldable display modules according to this embodiment in each group are the same, and other experimental conditions are also the same. Each experimental group represents a group of sample foldable display modules, and an experimental result of the group is an average value of experimental results of the group of sample foldable display modules.

TABLE-US-00003 TABLE 3 Group Control group Experimental Experimental Experimental 21 group 21 group 22 group 23 Structure Material Thickness Material Thickness Material Thickness Material Thickness Cover PET 50 PET 50 PET 50 PET 50 OCA 50 OCA 50 OCA 50 OCA 50 PET 100 TPU 150 TPU 150 TPU 150 OCA 50 OCA 50 OCA 50 OCA 50 PET 50 PET 100 OCA 50 OCA 50 Impact 2 cm ok 16 cm ok 30 cm ok 26 cm ok with a 32- 3 cm broken 18 cm broken 32 cm broken 28 cm broken g falling bright spots bright spots bright spots bright spots ball

[0082] It may be learned from the experimental data in Table 3 that, Control group 21 uses a PET layer without buffer performance, and a height of falling-ball impact resistance of the module is 2 cm. Experimental group 21 uses a TPU layer with buffer effect, and a height of falling-ball impact resistance of the module is 16 cm. For Experimental group 22 and Experimental group 23, high modulus PET (that is, the second film layer **14**) is added between TPU (that is, the buffer layer) and the flexible display, and heights of falling-ball impact resistance of the modules are 30 cm and 26 cm respectively. Therefore, in the screen cover **1** provided in this embodiment of this application, the buffer layer **11** is disposed to effectively reduce stress of impact, and reduce tensile strain on a TFT layer caused by the stress of impact, thereby improving impact resistance performance of the foldable display module. The second film layer **14** is added between the buffer layer **11** and the flexible display **2** to further improve the impact resistance performance of the foldable display module.

[0083] The foregoing descriptions are merely preferred embodiments of this application, and are not intended to limit this application. For a person skilled in the art, this application may have various modifications and variations. Any modification, equivalent replacement, or improvement made without departing from the spirit and principle of this application shall fall within the protection scope of this application.

Claims

1. A screen cover configured to cover a flexible display, wherein the screen cover comprises: a buffer layer comprising: a first side configured to be positioned away from the flexible display; and a second side configured to be positioned close to the flexible display, wherein the buffer layer is configured to absorb a stress of impact from a secondary object through elastic deformation; a first film layer covering the first side, wherein a first modulus of the first film layer is greater than a third modulus of the buffer; a first adhesive layer bonding the first film layer to the first side of the buffer layer; a second film layer covering the second side, wherein a second modulus of the second film layer is greater than the third modulus; and a second adhesive layer bonding the second film layer to the second side of the buffer layer.
2. The screen cover of claim 1, wherein the second film layer comprises a thickness of 25 micrometers (μm) to 150 μm .
3. The screen cover of claim 1, wherein the second film layer comprises a thickness of 25 micrometers (μm) to 75 μm .
4. The screen cover of claim 1, wherein the second modulus is not less than 1 gigapascal (GPa).
5. The screen cover of claim 1, wherein a material of the second film layer comprises colorless polyimide, polyester, epoxy resin, or ultra thin glass.
6. The screen cover of claim 1, wherein the second film layer and the first film layer are made of a same material.
7. A foldable display-module, apparatus comprising: a flexible display comprising a flexible display surface; and a screen cover covering the flexible display surface, wherein the screen cover comprises: a buffer layer comprising: a first side away from the flexible display; and a second side close to the flexible display, wherein the buffer layer is configured to absorb stress of impact through elastic deformation; a first film layer covering the first side, wherein a first modulus of the first film layer is greater than a third modulus of the buffer layer; a first adhesive layer bonding the

first film layer to the first side of the buffer layer; a second film layer covering the second side, wherein a second modulus of the second film layer is greater than the third modulus; and a second adhesive layer bonding the second film layer to the second side the buffer layer

8. A foldable display-module, apparatus comprising: a flexible display comprising a flexible display surface; and a screen cover covering a surface of the flexible display surface, wherein the screen cover comprises: a buffer layer having a buffer layer thickness comprising a first side away from the flexible display, and configured to absorb stress of impact from a secondary object through elastic deformation; a third adhesive layer bonding the buffer layer to the flexible display surface, wherein the third adhesive layer comprises a third adhesive layer thickness that is not less than half of the buffer layer thickness; a first film layer covering the first side, wherein a first modulus of the first film layer is greater than a second modulus of the buffer layer; and a first adhesive layer bonding the first film layer to the buffer layer.

9. The foldable display apparatus of claim 8, wherein the buffer layer thickness is not greater than 50 micrometers (μm) and the third adhesive layer thickness is 15 μm to 75 μm .

10. The foldable display apparatus of claim 8, wherein the buffer layer thickness is 50 micrometers (μm) to 100 μm and the third adhesive layer thickness is 25 μm to 100 μm .

11. The foldable display apparatus of claim 8, wherein the buffer layer thickness is 100 micrometers (μm) to 150 μm and the third adhesive layer thickness is 50 μm to 125 μm .

12. The foldable display apparatus of claim 8, wherein the buffer layer thickness is 150 micrometers (μm) to 200 μm and the third adhesive layer thickness is 50 μm to 150 μm .

13. The foldable display apparatus of claim 8, wherein the buffer layer thickness is not less than 200 micrometers (μm) and the third adhesive layer thickness is not less than 75 μm .

14. An electronic device, comprising: a flexible display comprising a flexible display surface; and a screen cover covering the flexible display surface, wherein the screen cover comprises: buffer layer having a buffer layer thickness, comprising a first side away from the flexible display, and configured to absorb stress of impact from a secondary object through elastic deformation; a third adhesive layer bonding the buffer layer to the flexible display surface, wherein the third adhesive layer comprises a third adhesive layer thickness that is not less than half of the buffer layer thickness; a first film layer covering the first side, wherein a first modulus of the first film layer is greater than a second modulus of the buffer layer; and a first adhesive layer bonding the first film layer to the buffer layer.

15. The electronic device of claim 14, wherein the buffer layer thickness is 150 micrometers (μm) to 200 μm , and the third adhesive layer thickness is 50 μm to 150 μm .

16. The electronic device of claim 14, wherein the buffer layer thickness is not greater than 50 micrometers (μm) and the third adhesive layer thickness is 15 μm to 75 μm .

17. The electronic device of claim 14, wherein the buffer layer thickness is 50 micrometers (μm) to 100 μm and the third adhesive layer thickness is 25 μm to 100 μm .

18. The electronic device of claim 14, wherein the buffer layer thickness is 100 micrometers (μm) to 150 μm and the third adhesive layer thickness is 50 μm to 125 μm .

19. The foldable display apparatus of claim 7, wherein the second film layer comprises a thickness of 25 micrometers (μm) to 150 μm .

20. The foldable display apparatus of claim 7, wherein the second film layer comprises a thickness of 25 micrometers (μm) to 75 μm .
